

sg13g2_io_slow_1p08V_3p0V_125C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx



*sg13g2_io_slow_1p08V_3p0V_125C Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C, Voltage 1.08, Temp
125.00*

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.02524	0.02443	0.30434	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.02523	0.02443	0.39755	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.02523	0.02443	0.22410	0.24000	12.00000
sg13g2_IOPadOut16mA	0.03679	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.03679	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.03679	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	707.25900	6093.75656	11193.60000
sg13g2_IOPadInOut30mA	787.16500	7156.66333	11193.70000
sg13g2_IOPadInOut4mA	717.83600	5232.29333	11193.60000
sg13g2_IOPadOut16mA	1097.26000	2867.41833	4977.35000
sg13g2_IOPadOut30mA	1519.28000	3999.05833	9082.49000
sg13g2_IOPadOut4mA	783.02500	1924.55917	3850.20000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (RR)	0.12000	0.02400	0.31378	0.80000	0.14400	0.57791	5.00000	0.24000	1.02892
sg13g2_IOPadInOut30mA	pad->p2c (RR)	0.12000	0.02400	0.31364	0.80000	0.14400	0.57790	5.00000	0.24000	1.02904
sg13g2_IOPadInOut4mA	pad->p2c (RR)	0.12000	0.02400	0.31393	0.80000	0.14400	0.57791	5.00000	0.24000	1.02883

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (FF)	0.12000	0.02400	0.31169	0.80000	0.14400	0.54345	5.00000	0.24000	0.88861
sg13g2_IOPadInOut30mA	pad->p2c (FF)	0.12000	0.02400	0.31162	0.80000	0.14400	0.54367	5.00000	0.24000	0.88885
sg13g2_IOPadInOut4mA	pad->p2c (FF)	0.12000	0.02400	0.31148	0.80000	0.14400	0.54321	5.00000	0.24000	0.88834

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (RR)	0.02000	1.00000	2.70572	0.33000	9.00000	4.26280	2.50000	20.00000	6.26136
	c2p_en->pad (FR)	0.02000	1.00000	2.05334	0.33000	9.00000	2.13891	2.50000	20.00000	2.38173
	c2p_en->pad (RR)	0.02000	1.00000	2.69373	0.33000	9.00000	4.28254	2.50000	20.00000	6.31537
sg13g2_IOPadInOut30mA	c2p->pad (RR)	0.02000	2.00000	3.19353	0.33000	18.00000	5.06366	2.50000	40.00000	7.21424
	c2p_en->pad (FR)	0.02000	2.00000	2.30789	0.33000	18.00000	2.39111	2.50000	40.00000	2.64161
	c2p_en->pad (RR)	0.02000	2.00000	3.17082	0.33000	18.00000	5.07661	2.50000	40.00000	7.23123
sg13g2_IOPadInOut4mA	c2p->pad (RR)	0.02000	1.00000	2.77925	0.33000	5.00000	5.51363	2.50000	12.00000	10.33310
	c2p_en->pad (FR)	0.02000	1.00000	1.79010	0.33000	5.00000	1.87319	2.50000	12.00000	2.12165
	c2p_en->pad (RR)	0.02000	1.00000	2.78549	0.33000	5.00000	5.58893	2.50000	12.00000	10.52770
sg13g2_IOPadOut16mA	c2p->pad (RR)	0.02000	1.00000	2.58932	0.33000	9.00000	4.20876	2.50000	20.00000	7.17701
sg13g2_IOPadOut30mA	c2p->pad (RR)	0.02000	2.00000	3.08351	0.33000	18.00000	5.01343	2.50000	40.00000	8.13582
sg13g2_IOPadOut4mA	c2p->pad (RR)	0.02000	1.00000	2.65018	0.33000	5.00000	5.44421	2.50000	12.00000	11.22390

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	2.73384	0.33000	9.00000	4.13686	2.50000	20.00000	6.30061
	c2p_en->pad (FF)	0.02000	1.00000	2.61954	0.33000	9.00000	2.71417	2.50000	20.00000	3.08801
	c2p_en->pad (RF)	0.02000	1.00000	2.50361	0.33000	9.00000	4.01403	2.50000	20.00000	5.99465
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	3.26578	0.33000	18.00000	4.83115	2.50000	40.00000	7.06549
	c2p_en->pad (FF)	0.02000	2.00000	3.47644	0.33000	18.00000	3.54153	2.50000	40.00000	3.94978
	c2p_en->pad (RF)	0.02000	2.00000	2.76150	0.33000	18.00000	4.48382	2.50000	40.00000	6.58798
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	2.87724	0.33000	5.00000	5.60304	2.50000	12.00000	10.60790
	c2p_en->pad (FF)	0.02000	1.00000	1.93720	0.33000	5.00000	2.02974	2.50000	12.00000	2.39469
	c2p_en->pad (RF)	0.02000	1.00000	2.87660	0.33000	5.00000	5.68033	2.50000	12.00000	10.59360
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	2.58499	0.33000	9.00000	3.91089	2.50000	20.00000	5.83027
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	3.12490	0.33000	18.00000	4.62090	2.50000	40.00000	6.60126
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	2.69037	0.33000	5.00000	5.34805	2.50000	12.00000	10.10970

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	-1.12650	0.80000	0.14400	-1.07916	5.00000	0.24000	-0.69998
	pad	0.12000	0.02400	0.02107	0.80000	0.14400	0.02220	5.00000	0.24000	0.02253
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	-1.75961	0.80000	0.14400	-1.70778	5.00000	0.24000	-1.33049
	pad	0.12000	0.02400	0.02124	0.80000	0.14400	0.02239	5.00000	0.24000	0.02275
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	-0.61205	0.80000	0.14400	-0.56797	5.00000	0.24000	-0.18490
	pad	0.12000	0.02400	0.02096	0.80000	0.14400	0.02211	5.00000	0.24000	0.02240

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	1.49813	0.80000	0.14400	1.53974	5.00000	0.24000	1.91644
	pad	0.12000	0.02400	0.05061	0.80000	0.14400	0.05426	5.00000	0.24000	0.07067
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	2.13047	0.80000	0.14400	2.16861	5.00000	0.24000	2.54802
	pad	0.12000	0.02400	0.05042	0.80000	0.14400	0.05401	5.00000	0.24000	0.07044
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	0.98471	0.80000	0.14400	1.02855	5.00000	0.24000	1.40503
	pad	0.12000	0.02400	0.05055	0.80000	0.14400	0.05442	5.00000	0.24000	0.07083

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	5.33132	0.33000	9.00000	5.22828	2.50000	20.00000	4.97706
	c2p	0.02000	1.00000	1.21793	0.33000	9.00000	10.47570	2.50000	20.00000	10.66660
	c2p_en	0.02000	1.00000	4.91585	0.33000	9.00000	5.67255	2.50000	20.00000	6.62541
	c2p_en	0.02000	1.00000	1.19612	0.33000	9.00000	10.45690	2.50000	20.00000	10.65730
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	8.03659	0.33000	18.00000	7.64916	2.50000	40.00000	6.99912
	c2p	0.02000	2.00000	2.37060	0.33000	18.00000	16.35940	2.50000	40.00000	0.06959
	c2p_en	0.02000	2.00000	7.27464	0.33000	18.00000	8.07450	2.50000	40.00000	8.44992
	c2p_en	0.02000	2.00000	2.35188	0.33000	18.00000	16.33580	2.50000	40.00000	0.05232
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	3.13885	0.33000	5.00000	3.52821	2.50000	12.00000	3.79959
	c2p	0.02000	1.00000	1.21802	0.33000	5.00000	5.86872	2.50000	12.00000	13.98720
	c2p_en	0.02000	1.00000	2.88449	0.33000	5.00000	3.62406	2.50000	12.00000	4.98792
	c2p_en	0.02000	1.00000	1.19485	0.33000	5.00000	5.79963	2.50000	12.00000	13.97120
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	4.99094	0.33000	9.00000	4.88019	2.50000	20.00000	4.68354
	c2p	0.02000	1.00000	-0.01174	0.33000	9.00000	-0.01294	2.50000	20.00000	0.01343
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	7.83292	0.33000	18.00000	7.52259	2.50000	40.00000	7.11390
	c2p	0.02000	2.00000	-0.01174	0.33000	18.00000	-0.01294	2.50000	40.00000	0.01346
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.67546	0.33000	5.00000	2.57910	2.50000	12.00000	2.49603
	c2p	0.02000	1.00000	-0.01174	0.33000	5.00000	-0.01294	2.50000	12.00000	0.01367

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	10.77700	0.33000	9.00000	7.41003	2.50000	20.00000	6.60234
	c2p	0.02000	1.00000	0.14922	0.33000	9.00000	0.15566	2.50000	20.00000	0.18463
	c2p_en	0.02000	1.00000	3.05627	0.33000	9.00000	3.23627	2.50000	20.00000	3.52985
	c2p_en	0.02000	1.00000	0.08262	0.33000	9.00000	0.09069	2.50000	20.00000	0.11367
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	38.37660	0.33000	18.00000	23.46430	2.50000	40.00000	18.08520
	c2p	0.02000	2.00000	0.15050	0.33000	18.00000	0.15618	2.50000	40.00000	0.17918
	c2p_en	0.02000	2.00000	4.52848	0.33000	18.00000	4.72207	2.50000	40.00000	5.02880
	c2p_en	0.02000	2.00000	0.08313	0.33000	18.00000	0.09086	2.50000	40.00000	0.10801
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	2.55594	0.33000	5.00000	2.97188	2.50000	12.00000	3.72879
	c2p	0.02000	1.00000	0.15344	0.33000	5.00000	0.16762	2.50000	12.00000	0.20903
	c2p_en	0.02000	1.00000	1.91053	0.33000	5.00000	2.32903	2.50000	12.00000	3.08578
	c2p_en	0.02000	1.00000	0.08734	0.33000	5.00000	0.10216	2.50000	12.00000	0.13727
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	11.60990	0.33000	9.00000	7.35884	2.50000	20.00000	6.06470
	c2p	0.02000	1.00000	0.04894	0.33000	9.00000	0.04923	2.50000	20.00000	0.07471
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	40.58500	0.33000	18.00000	24.27490	2.50000	40.00000	18.12080
	c2p	0.02000	2.00000	0.04895	0.33000	18.00000	0.04921	2.50000	40.00000	0.07477
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.23282	0.33000	5.00000	2.15332	2.50000	12.00000	2.11604
	c2p	0.02000	1.00000	0.04895	0.33000	5.00000	0.04922	2.50000	12.00000	0.07765

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	-0.00007	0.33000	-0.00005	2.50000	-0.00005
	0.02000	-0.01390	0.33000	-0.01469	2.50000	-0.01495
sg13g2_IOPadInOut30mA	0.02000	-0.00008	0.33000	-0.00005	2.50000	-0.00005
	0.02000	-0.01391	0.33000	-0.01469	2.50000	-0.01495
sg13g2_IOPadInOut4mA	0.02000	-0.00008	0.33000	-0.00005	2.50000	-0.00005
	0.02000	-0.01390	0.33000	-0.01469	2.50000	-0.01495

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.00007	0.33000	0.00005	2.50000	0.00005
	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
sg13g2_IOPadInOut30mA	0.02000	0.00008	0.33000	0.00005	2.50000	0.00005
	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
sg13g2_IOPadInOut4mA	0.02000	0.00008	0.33000	0.00005	2.50000	0.00005
	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	-0.00007	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * pad * p2c)	0.02000	-0.01390	0.33000	-0.01468	2.50000	-0.01495
	(!c2p_en * !pad * !p2c)	0.02000	-0.00007	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * !pad * !p2c)	0.02000	-0.01390	0.33000	-0.01469	2.50000	-0.01495
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	-0.00007	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * pad * p2c)	0.02000	-0.01391	0.33000	-0.01468	2.50000	-0.01495
	(!c2p_en * !pad * !p2c)	0.02000	-0.00008	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * !pad * !p2c)	0.02000	-0.01391	0.33000	-0.01469	2.50000	-0.01495
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	-0.00007	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * pad * p2c)	0.02000	-0.01390	0.33000	-0.01468	2.50000	-0.01495
	(!c2p_en * !pad * !p2c)	0.02000	-0.00008	0.33000	-0.00005	2.50000	-0.00005
	(!c2p_en * !pad * !p2c)	0.02000	-0.01390	0.33000	-0.01469	2.50000	-0.01495

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	0.00007	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * pad * p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
	(!c2p_en * !pad * !p2c)	0.02000	0.00007	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * !pad * !p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	0.00007	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * pad * p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
	(!c2p_en * !pad * !p2c)	0.02000	0.00008	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * !pad * !p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	0.00007	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * pad * p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677
	(!c2p_en * !pad * !p2c)	0.02000	0.00008	0.33000	0.00005	2.50000	0.00005
	(!c2p_en * !pad * !p2c)	0.02000	0.01710	0.33000	0.01679	2.50000	0.01677

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	1.40117	0.33000	1.40078	2.50000	1.40542
	0.02000	0.02911	0.33000	0.02734	2.50000	0.04219
sg13g2_IOPadInOut30mA	0.02000	2.13578	0.33000	2.13520	2.50000	2.13910
	0.02000	0.02911	0.33000	0.02733	2.50000	0.04219
sg13g2_IOPadInOut4mA	0.02000	0.77890	0.33000	0.77877	2.50000	0.78282
	0.02000	0.02912	0.33000	0.02732	2.50000	0.04218

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.30845	0.33000	0.30783	2.50000	0.30636
	0.02000	0.06560	0.33000	0.06481	2.50000	0.07827
sg13g2_IOPadInOut30mA	0.02000	0.30135	0.33000	0.30073	2.50000	0.29915
	0.02000	0.06561	0.33000	0.06481	2.50000	0.07827
sg13g2_IOPadInOut4mA	0.02000	0.32664	0.33000	0.32602	2.50000	0.32487
	0.02000	0.06561	0.33000	0.06489	2.50000	0.07829

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	5.33132	0.33000	5.22828	2.50000	4.97706
	0.02000	1.21793	0.33000	10.47570	2.50000	10.66660
	0.02000	4.91585	0.33000	5.67255	2.50000	6.62541
	0.02000	1.19612	0.33000	10.45690	2.50000	10.65730
sg13g2_IOPadInOut30mA	0.02000	8.03659	0.33000	7.64916	2.50000	6.99912
	0.02000	2.37060	0.33000	16.35940	2.50000	0.06959
	0.02000	7.27464	0.33000	8.07450	2.50000	8.44992
	0.02000	2.35188	0.33000	16.33580	2.50000	0.05232
sg13g2_IOPadInOut4mA	0.02000	3.13885	0.33000	3.52821	2.50000	3.79959
	0.02000	1.21802	0.33000	5.86872	2.50000	13.98720
	0.02000	2.88449	0.33000	3.62406	2.50000	4.98792
	0.02000	1.19485	0.33000	5.79963	2.50000	13.97120
sg13g2_IOPadOut16mA	0.02000	4.99094	0.33000	4.88019	2.50000	4.68354
	0.02000	-0.01174	0.33000	-0.01294	2.50000	0.01343
sg13g2_IOPadOut30mA	0.02000	7.83292	0.33000	7.52259	2.50000	7.11390
	0.02000	-0.01174	0.33000	-0.01294	2.50000	0.01346
sg13g2_IOPadOut4mA	0.02000	2.67546	0.33000	2.57910	2.50000	2.49603
	0.02000	-0.01174	0.33000	-0.01294	2.50000	0.01367

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	10.77700	0.33000	7.41003	2.50000	6.60234
	0.02000	0.14922	0.33000	0.15566	2.50000	0.18463
	0.02000	3.05627	0.33000	3.23627	2.50000	3.52985
	0.02000	0.08262	0.33000	0.09069	2.50000	0.11367
sg13g2_IOPadInOut30mA	0.02000	38.37660	0.33000	23.46430	2.50000	18.08520
	0.02000	0.15050	0.33000	0.15618	2.50000	0.17918
	0.02000	4.52848	0.33000	4.72207	2.50000	5.02880
	0.02000	0.08313	0.33000	0.09086	2.50000	0.10801
sg13g2_IOPadInOut4mA	0.02000	2.55594	0.33000	2.97188	2.50000	3.72879
	0.02000	0.15344	0.33000	0.16762	2.50000	0.20903
	0.02000	1.91053	0.33000	2.32903	2.50000	3.08578
	0.02000	0.08734	0.33000	0.10216	2.50000	0.13727
sg13g2_IOPadOut16mA	0.02000	11.60990	0.33000	7.35884	2.50000	6.06470
	0.02000	0.04894	0.33000	0.04923	2.50000	0.07471
sg13g2_IOPadOut30mA	0.02000	40.58500	0.33000	24.27490	2.50000	18.12080
	0.02000	0.04895	0.33000	0.04921	2.50000	0.07477
sg13g2_IOPadOut4mA	0.02000	2.23282	0.33000	2.15332	2.50000	2.11604
	0.02000	0.04895	0.33000	0.04922	2.50000	0.07765

INPUT



*sg13g2_io_slow_1p08V_3p0V_125C Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C, Voltage 1.08, Temp 125.00*

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23620	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	369.32000	869.59533	1941.11000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (RR)	0.12000	0.02400	0.31348	0.80000	0.14400	0.57684	5.00000	0.24000	1.02728

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (FF)	0.12000	0.02400	0.31155	0.80000	0.14400	0.54256	5.00000	0.24000	0.88812

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	-0.53083	0.80000	0.14400	-0.48632	5.00000	0.24000	-0.10707
	pad	0.12000	0.02400	0.02148	0.80000	0.14400	0.02253	5.00000	0.24000	0.02299

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	0.90286	0.80000	0.14400	0.94588	5.00000	0.24000	1.32624
	pad	0.12000	0.02400	0.05001	0.80000	0.14400	0.05380	5.00000	0.24000	0.07020

SG13G2_CORNER



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG



sg13g2_io_slow_1p08V_3p0V_125C

Cell Library: Process

sg13g2_io_slow_1p08V_3p0V_125C,

Voltage 1.08, Temp 125.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.78661	0.59956	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	5820.60000	11641.20000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (R)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (F)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

SG13G2_IOPADIOVDD



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS



sg13g2_io_slow_1p08V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p08V_3p0V_125C,
Voltage 1.08, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx



*sg13g2_io_slow_1p08V_3p0V_125C Cell Library:
Process sg13g2_io_slow_1p08V_3p0V_125C, Voltage
1.08, Temp 125.00*

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.02519	0.02479	20.00000
sg13g2_IOPadTriOut30mA	0.02519	0.02479	40.00000
sg13g2_IOPadTriOut4mA	0.02519	0.02480	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	915.54900	4852.90970	9373.29000
sg13g2_IOPadTriOut30mA	1329.55000	5521.59600	9373.80000
sg13g2_IOPadTriOut4mA	568.89100	4286.54090	9373.34000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	2.70238	0.33000	9.00000	4.26481	2.50000	20.00000	6.26321
	c2p_en->pad (FR)	0.02000	1.00000	2.06613	0.33000	9.00000	2.15093	2.50000	20.00000	2.39316
	c2p_en->pad (RR)	0.02000	1.00000	2.69368	0.33000	9.00000	4.28530	2.50000	20.00000	6.31310
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	3.19560	0.33000	18.00000	5.06984	2.50000	40.00000	7.21971
	c2p_en->pad (FR)	0.02000	2.00000	2.32086	0.33000	18.00000	2.40445	2.50000	40.00000	2.65521
	c2p_en->pad (RR)	0.02000	2.00000	3.17321	0.33000	18.00000	5.07234	2.50000	40.00000	7.22857
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	2.76396	0.33000	5.00000	5.50058	2.50000	12.00000	10.31560
	c2p_en->pad (FR)	0.02000	1.00000	1.80215	0.33000	5.00000	1.88715	2.50000	12.00000	2.13671
	c2p_en->pad (RR)	0.02000	1.00000	2.77745	0.33000	5.00000	5.58405	2.50000	12.00000	10.52570

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	2.73897	0.33000	9.00000	4.14305	2.50000	20.00000	6.30863
	c2p_en->pad (FF)	0.02000	1.00000	2.64084	0.33000	9.00000	2.74042	2.50000	20.00000	3.10809
	c2p_en->pad (RF)	0.02000	1.00000	2.50986	0.33000	9.00000	4.02451	2.50000	20.00000	6.00355
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	3.27741	0.33000	18.00000	4.84420	2.50000	40.00000	7.07308
	c2p_en->pad (FF)	0.02000	2.00000	3.46088	0.33000	18.00000	3.60430	2.50000	40.00000	3.96469
	c2p_en->pad (RF)	0.02000	2.00000	2.76825	0.33000	18.00000	4.49727	2.50000	40.00000	6.60052
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	2.86856	0.33000	5.00000	5.59383	2.50000	12.00000	10.59710
	c2p_en->pad (FF)	0.02000	1.00000	1.95135	0.33000	5.00000	2.04307	2.50000	12.00000	2.40791
	c2p_en->pad (RF)	0.02000	1.00000	2.87120	0.33000	5.00000	5.66962	2.50000	12.00000	10.57110

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	4.96282	0.33000	9.00000	4.87989	2.50000	20.00000	4.61505
	c2p	0.02000	1.00000	0.04242	0.33000	9.00000	0.04016	2.50000	20.00000	0.06151
	c2p_en	0.02000	1.00000	4.49681	0.33000	9.00000	5.23863	2.50000	20.00000	6.02640
	c2p_en	0.02000	1.00000	0.01914	0.33000	9.00000	0.01745	2.50000	20.00000	0.04610
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	7.76423	0.33000	18.00000	7.51002	2.50000	40.00000	7.65840
	c2p	0.02000	2.00000	0.04243	0.33000	18.00000	0.04015	2.50000	40.00000	0.06151
	c2p_en	0.02000	2.00000	6.89637	0.33000	18.00000	7.41477	2.50000	40.00000	7.96170
	c2p_en	0.02000	2.00000	0.01914	0.33000	18.00000	0.01745	2.50000	40.00000	0.04611
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.67175	0.33000	5.00000	2.57208	2.50000	12.00000	2.46051
	c2p	0.02000	1.00000	0.04247	0.33000	5.00000	0.04021	2.50000	12.00000	0.06159
	c2p_en	0.02000	1.00000	2.41188	0.33000	5.00000	2.84337	2.50000	12.00000	3.66888
	c2p_en	0.02000	1.00000	0.01920	0.33000	5.00000	0.01749	2.50000	12.00000	0.04545

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	10.45410	0.33000	9.00000	6.90778	2.50000	20.00000	5.82094
	c2p	0.02000	1.00000	0.09500	0.33000	9.00000	0.09247	2.50000	20.00000	0.11368
	c2p_en	0.02000	1.00000	2.74301	0.33000	9.00000	2.74138	2.50000	20.00000	2.74289
	c2p_en	0.02000	1.00000	0.02914	0.33000	9.00000	0.02742	2.50000	20.00000	0.04228
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	37.94570	0.33000	18.00000	22.96420	2.50000	40.00000	17.20670
	c2p	0.02000	2.00000	0.09501	0.33000	18.00000	0.09248	2.50000	40.00000	0.11369
	c2p_en	0.02000	2.00000	4.20414	0.33000	18.00000	4.20671	2.50000	40.00000	4.20734
	c2p_en	0.02000	2.00000	0.02916	0.33000	18.00000	0.02743	2.50000	40.00000	0.04228
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.17885	0.33000	5.00000	2.17749	2.50000	12.00000	2.18240
	c2p	0.02000	1.00000	0.09516	0.33000	5.00000	0.09263	2.50000	12.00000	0.11386
	c2p_en	0.02000	1.00000	1.52191	0.33000	5.00000	1.52149	2.50000	12.00000	1.52499
	c2p_en	0.02000	1.00000	0.02932	0.33000	5.00000	0.02757	2.50000	12.00000	0.04210

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.01701	0.33000	0.01663	2.50000	0.01675
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.01702	0.33000	0.01664	2.50000	0.01675
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.01700	0.33000	0.01663	2.50000	0.01675

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.01386	0.33000	-0.01477	2.50000	-0.01495

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.01701	0.33000	0.01663	2.50000	0.01675
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.01702	0.33000	0.01664	2.50000	0.01675
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.01700	0.33000	0.01663	2.50000	0.01675